Performance Evaluation

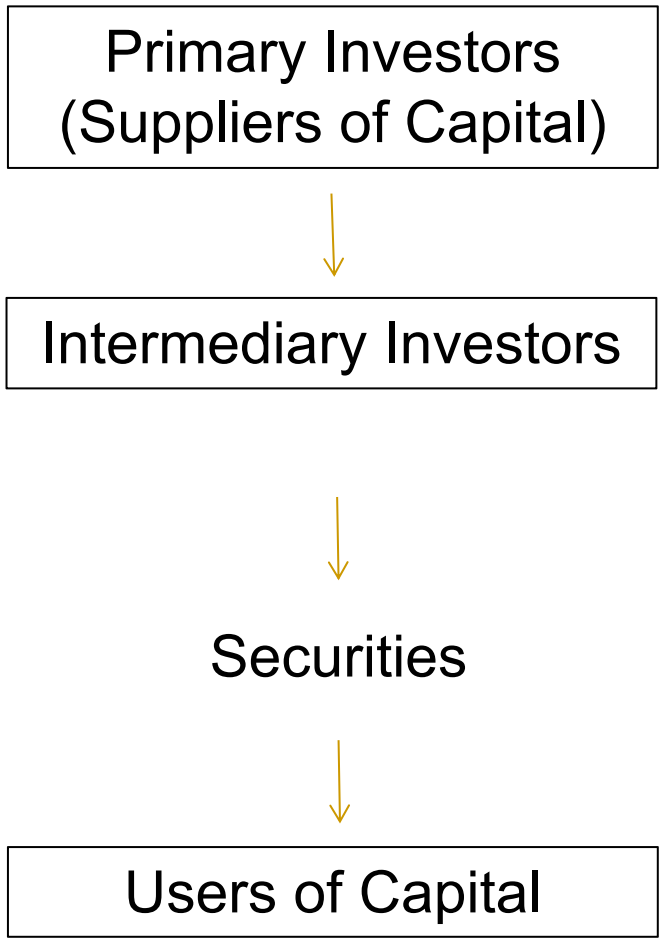
Jinfan Zhang
CUHK(Shenzhen)

Overview

- Last time:
  - Defined market efficiency
  - Looked at some evidence for and against market efficiency
  - Examined arguments by the behavioral and rational camps

- Today: Money management industry
  - Structure and key players
  - Their performance

The Financial System



- Households with savings
- Firms with cash
- Endowments
- Sovereign wealth funds

- Mutual funds
- Hedge funds
- Insurance companies
- Private equity funds
- Banks

- Government bonds
- Corporate bonds
- Equities

- Firms which want to invest
- Governments who want to spend

Review various types of intermediary investors

Mutual Funds

- Pool money from many investors and invest that money in stocks, bonds, or other securities
  - cater to individual investors
    为...服务
- Charge a fee equal to some fraction of assets under management (AUM)
  - e.g. Fidelity; Vanguard

Hedge Funds

- Pool money from investors and invest that money in stocks, bonds, or other securities
- Differ from mutual funds in that they use more aggressive investment techniques
  - short-selling, leverage, derivatives
- Limited to “sophisticated” primary investors (pension plans, endowments, or wealthy individuals)
- Charge a fee that is some fraction of AUM, and also collect some fraction of the fund’s profits
  - e.g. Citadel, SAC Capital, LTCM

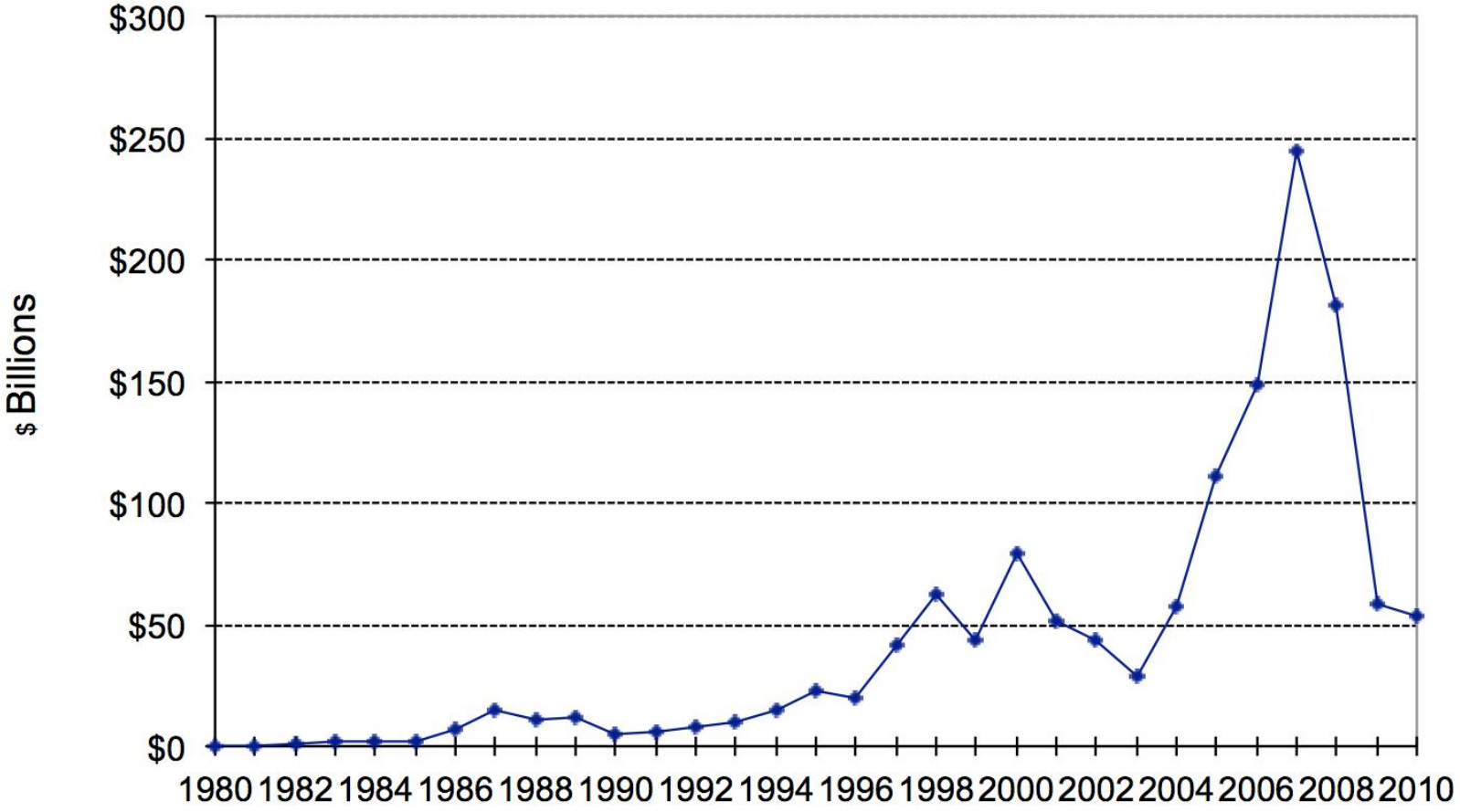
Private Equity Funds (many different types)

- Venture capital and PE funds
  - Pool money from primary investors and use it to invest in early-stage firms
  - exit the investment a few years later, typically in an IPO
  - Charge a fee that is some fraction of AUM, and also collect some fraction of the fund's profits

Private Equity Funds (many different types)

- Leveraged buyout funds (LBO funds) 收购
  - Pool money from primary investors and use it, in conjunction with significant borrowing, to buy entire companies
  - exit from the investment a few years later
  - Charge a fee that is some fraction of AUM, and also collect some fraction of the fund's profits
  - e.g. KKR, TPG Capital, Bain Capital, Blackstone

Commitments to U.S. Private Equity Partnerships 1980 - 2010 (in $ billions)



Source: Private Equity Analyst, Steven N. Kaplan

Average Performance

How to measure?

- Look at their average returns, both before and after fees

- Look at whether they beat benchmarks
  - index benchmarks: S&P 500, Wilshire 5000
  - CAPM benchmark

The CAPM benchmark

- Run a regression of excess fund returns on excess market returns:

$$r_{P,t} - r_{f,t} = \alpha_P + \beta_P(r_{M,t} - r_{f,t}) + \epsilon_{P,t}$$

- If $\alpha_P > 0$, this is interpreted as evidence that the manager has skill

- The fund's "alpha"

$$\underbrace{\alpha_P}_{\text{Skill}} = \underbrace{\bar{r}_p}_{\text{How the fund did}} - \underbrace{\left(\bar{r}_f + \beta_P\left(\overline{r_M - r_f}\right)\right)}_{\text{How the fund should have done, given the risk}}$$

Institutional Fund Managers

- Poor record. Studies have found underperformance, relative to benchmarks, even before fees!

Table 2. Annual Return of Equity Funds and Percentage Underperforming *S&P 500*, 1983–89[a]

Percent

Year	*Equally-weighted return across funds*	*Value-weighted return across funds*	S&P 500 *return*	*Funds underperforming* S&P 500
1983	17.8	18.1	22.5	59
1984	3.8	3.2	6.3	63
1985	33.3	30.5	32.2	38
1986	18.1	16.8	18.5	50
1987	4.0	4.4	5.2	61
1988	17.9	15.7	16.8	47
1989	29.2	25.9	31.5	61
Mean across years	17.7	16.4	19.0	54

a. Performance database excluding cash portfolio.

Source: Lakonishok, Shleifer, and Vishny (1992)

Institutional Fund Managers

- In more recent studies
  - In US stock market, during the period of 1971-2009, the average annual return of actively managed stock mutual funds is 1% lower than the return of Wilshire 5000 Index. The average returns lose to the index returns in 23 years out of 29 years.

Mutual Funds and Hedge Funds

- Mutual Funds
  - On average, can not beat the market even before the fee
- Hedge Funds:
  - Outperform benchmarks slightly, before fees, but underperform, after fees

Private Equity Funds

- Private equity funds:
  - On average, both LBO funds and venture capital funds outperform the S&P 500 slightly, before fees, but underperform, after fees
  - Particularly unimpressive, given the illiquidity of the investments, and the use of leverage
  - Different types of investors have different returns

Are some funds persistently better than others?

Consistency of performance of pension funds

Table 9. Equity Fund Performance over Time Using Quartile Rankings of One-Year Past Performance to Predict Future One-Year Performance[a]

Investment style		*1[b]*	*2[b]*	*3[b]*	*4[b]*	*Return over past one-year period[c]*	*Return over future one-year period[c]*
All							
(Top)	1	26	24	23	27	25.5	17.6
	2	20	26	29	25	18.4	17.3
	3	22	28	26	24	13.7	17.4
(Bottom)	4	32	22	22	24	5.5	18.5

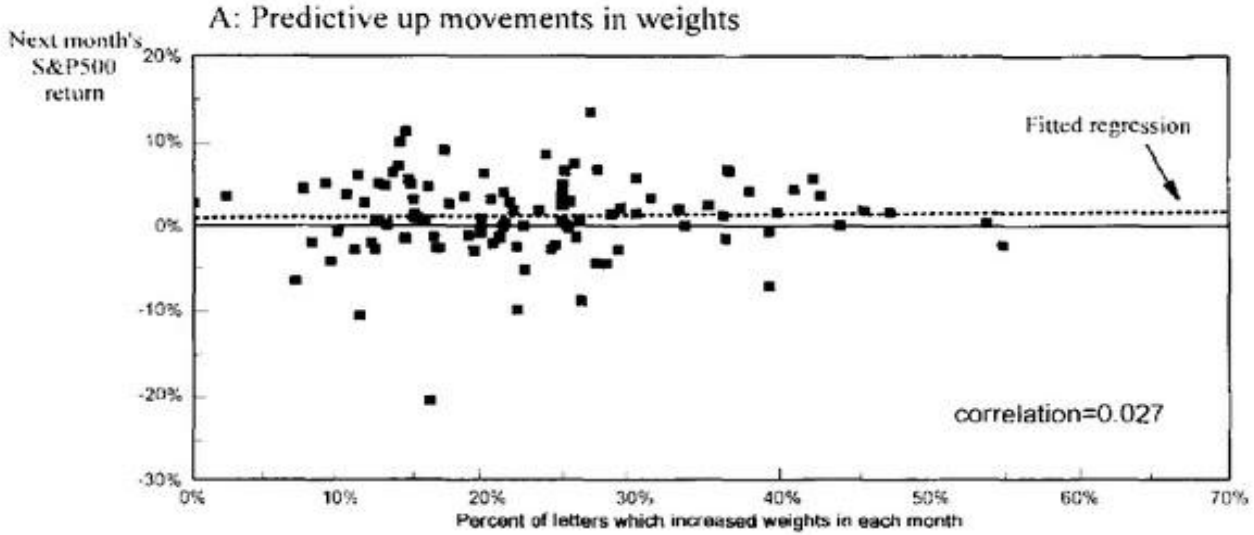
Source: Lakonishok, Shleifer, and Vishny (1992)

Consistency of performance of others

- Mutual funds
  - Some limited persistence from year to year
- Hedge funds
  - Some limited persistence from year to year
- Private equity funds
  - Quite strong persistence from year to year, but it could due to the valuation bias

Can mutual fund managers beat market?

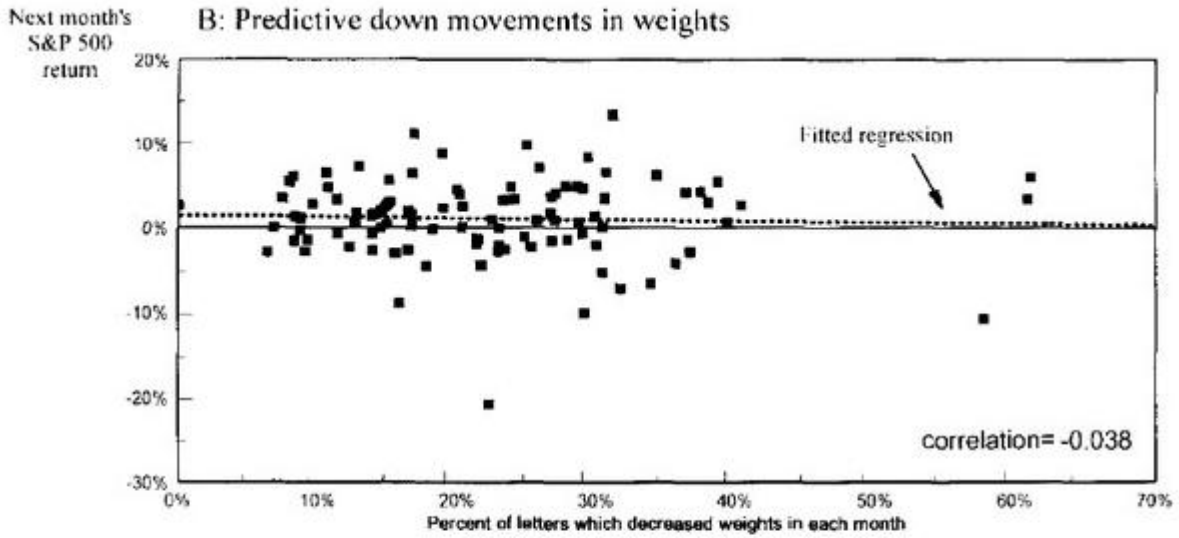
- The recommendations of Investment magazine can not predict the stock market movement
  - When they recommend buying stocks



Source: Graham, John R., and Campbell R. Harvey. &# 34;Market timing ability and volatility implied in investment newsletters&# 39; asset allocation recommendations.&# 34; Journal of Financial Economics 42.3 (1996): 397-421.

Can mutual fund managers beat market?

- The recommendations of Investment magazine can not predict the stock market movement
  - When they recommend selling stocks



Source: Graham, John R., and Campbell R. Harvey. &# 34;Market timing ability and volatility implied in investment newsletters&# 39; asset allocation recommendations.&# 34; Journal of Financial Economics 42.3 (1996): 397-421.

Implications on market efficiency

- Overall, intermediary investors do not seem particularly skilled at exploiting mispricing
  - before fees, intermediary investors do not significantly outperform benchmarks
  - little persistence in fund returns
  - even existence of people like Peter Lynch does not prove that stock-picking ability exists (e.g., coin-flip example)

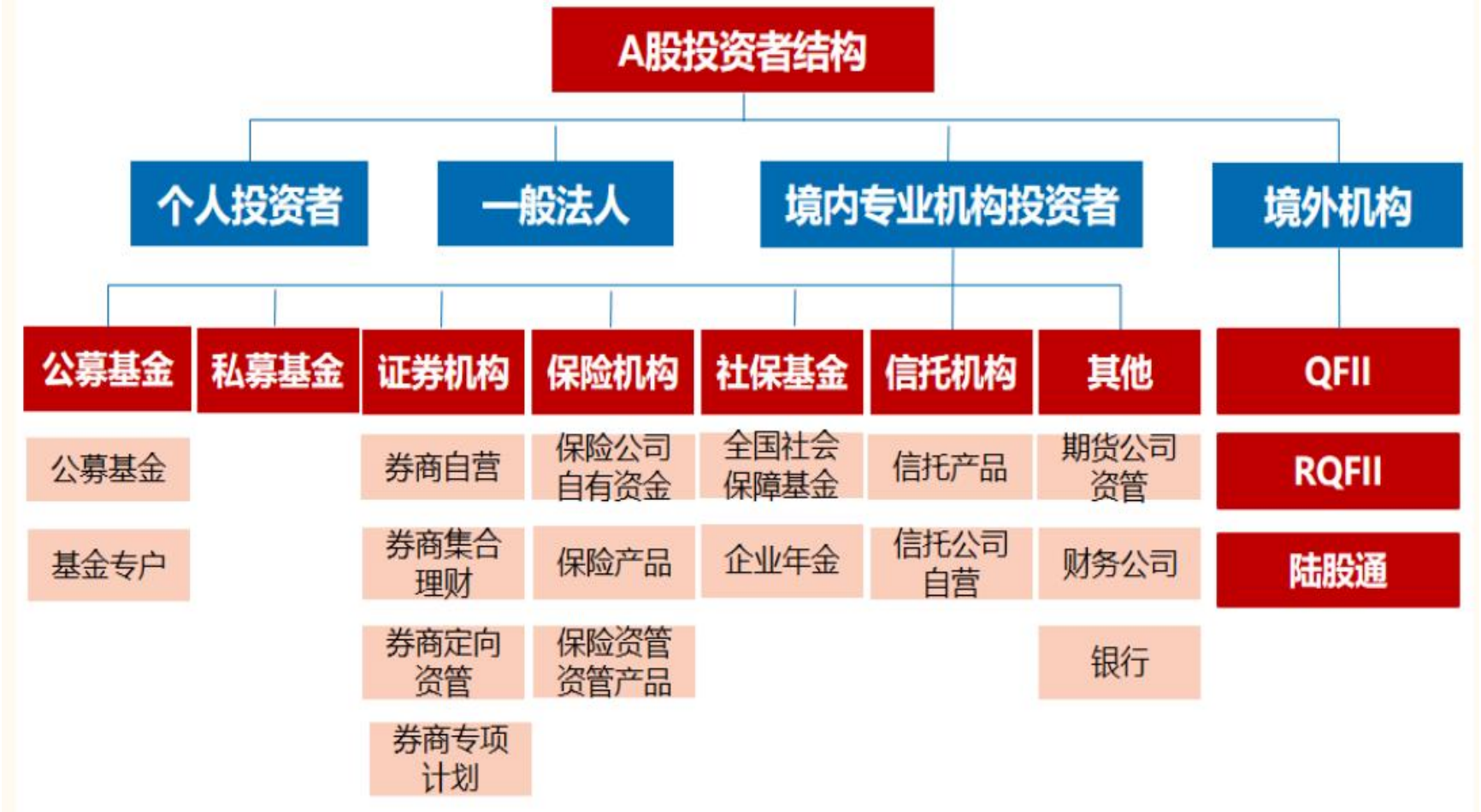
Implications on market efficiency

- But the weak performance of intermediary investors before fees *can* still be consistent with inefficient markets
  - Even if prices are wrong, it's not easy to beat the market
  - Intermediary investors may be swamped with more money than they can usefully invest, lowering their average returns
  - Selection bias: outstanding investors do not want to disclose themselves (hedge fund)

Implications for primary investors

- Investors
  - even though markets probably contain inefficiencies, index funds are still best for individuals
  - active money managers don't beat benchmarks, after fees
  - it's puzzling why actively managed mutual funds remain so popular

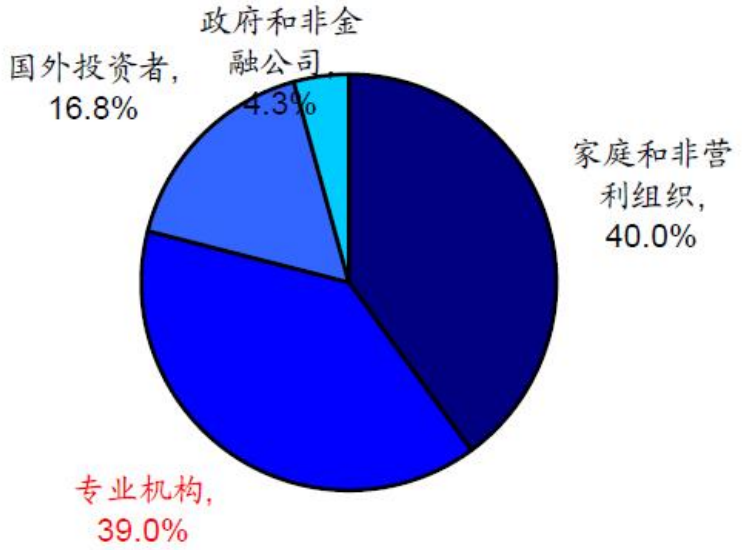
Investor Structure



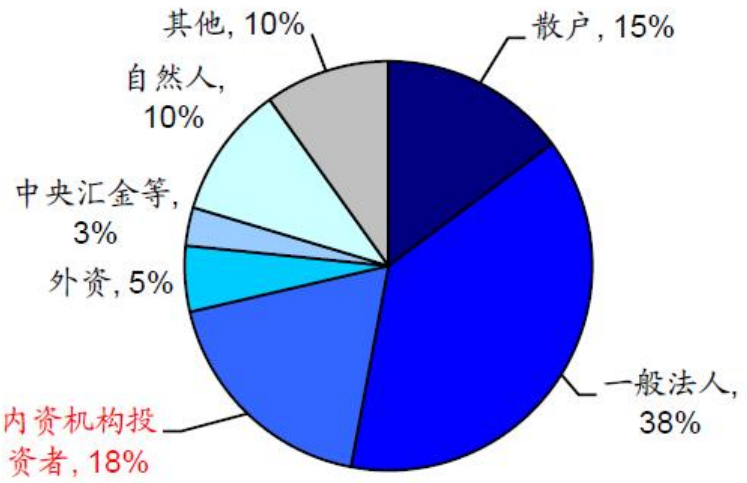
Investor Structure

- Institution holding (US 39%; China 18%)

2022 年 2 季度美国股票市场投资者结构



2022 年 3 季度中国股票市场投资者结构

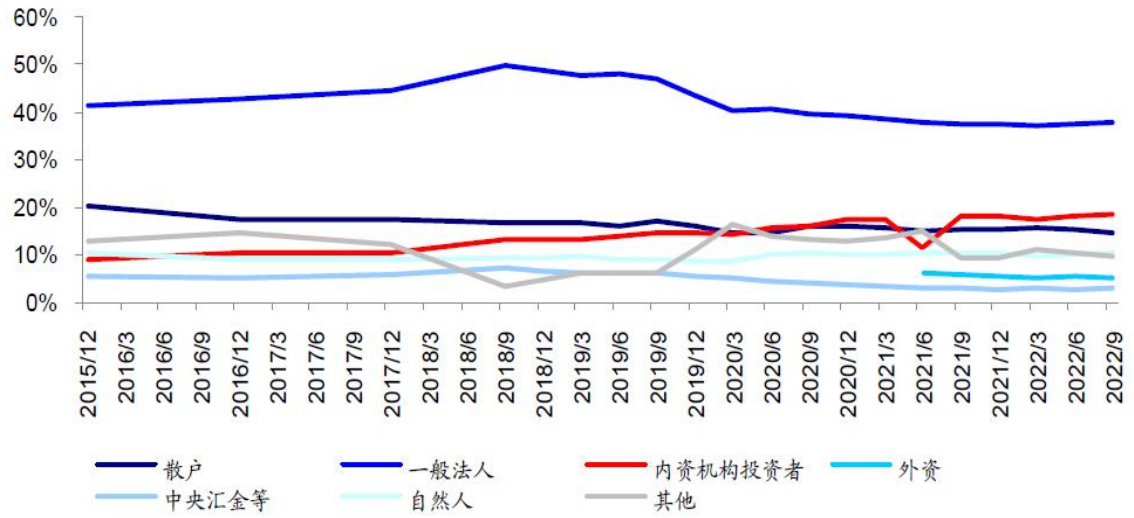


From 2019 to 2021年，retail investor's trading volume accounts for 78.00%、72.60% and 65.10%. Institutional investor's trading volume rise from 12.43% in 2018 to 28.85% in 2022. Foreign investor's trading volume rises from 3.49% in 2018 to 9.08% in 2022.

Institutionization

- From 2019Q1 - 2022Q3
  - 公募基金持股市值: （Mutual Fund Holding）
    - 1.95万亿元→5.3万亿元，占比4.3%→8.3%；
  - 保险（含社保、年金） （Insurance Company Holding）
    - 1.92万亿元→2.8万亿，占比从4.2%→ 4.4%；
  - 私募 （Hedge Fund Holding）
    - 0.8万亿元→2.4万亿元，占比从1.9%→3.8%；
  - 外资 （Foreign Investors）
    - 1.68万亿元→3万亿元，占比从3.7%→4.8%。

图1 2015Q4-2022Q3 A 股投资者流通市值占比（总市值口径）



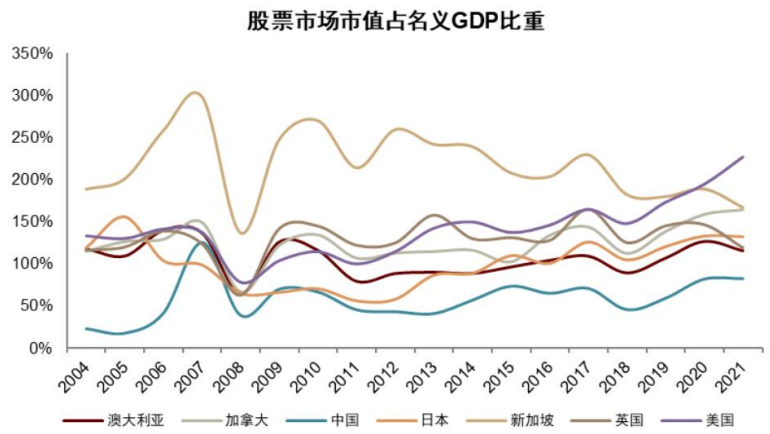
Institutionization



From 2019 to 2022, retail investors' trading volume account for 78.00%、72.60%, 65.10% and 61.35%. The trading volume of domestic mutual funds and hedge funds goes to 28.85% in 2022 (compared to 12.43% in 2018)，the trading volume of foreigners grow from 3.49% to 9.08% in the same period.

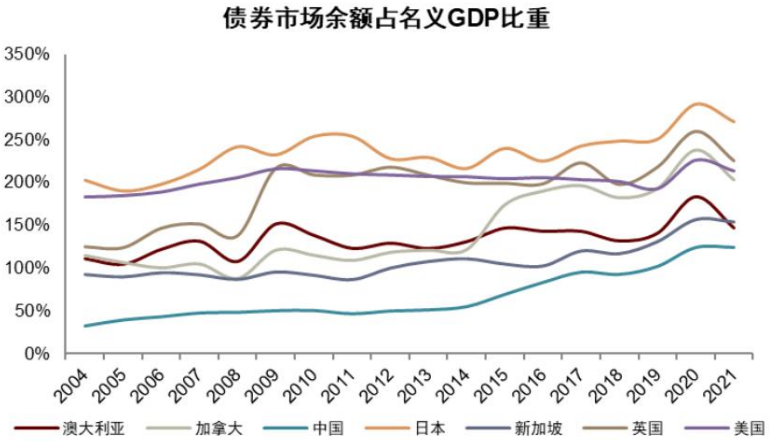
Great Potential of China's Financial Market

Stock Mkt Cap/GDP



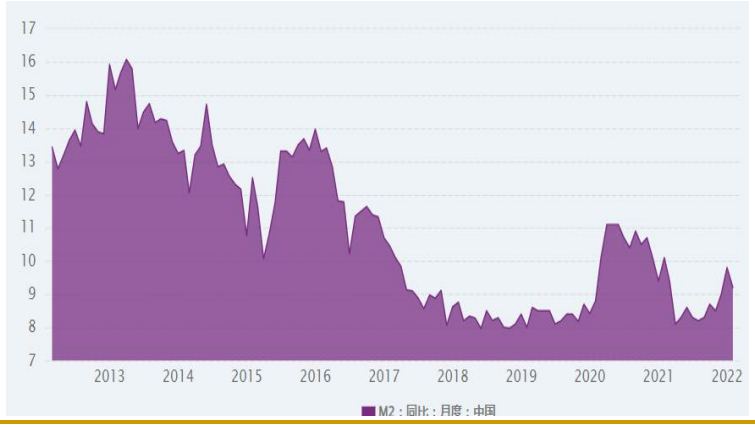
Australia Canada China Japan Singapore England US

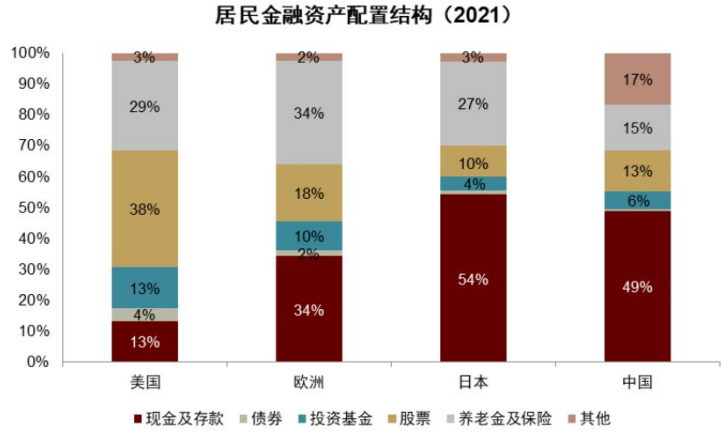
Bond Mkt Cap/GDP



Australia Canada China Japan Singapore England US

M2 growth rate



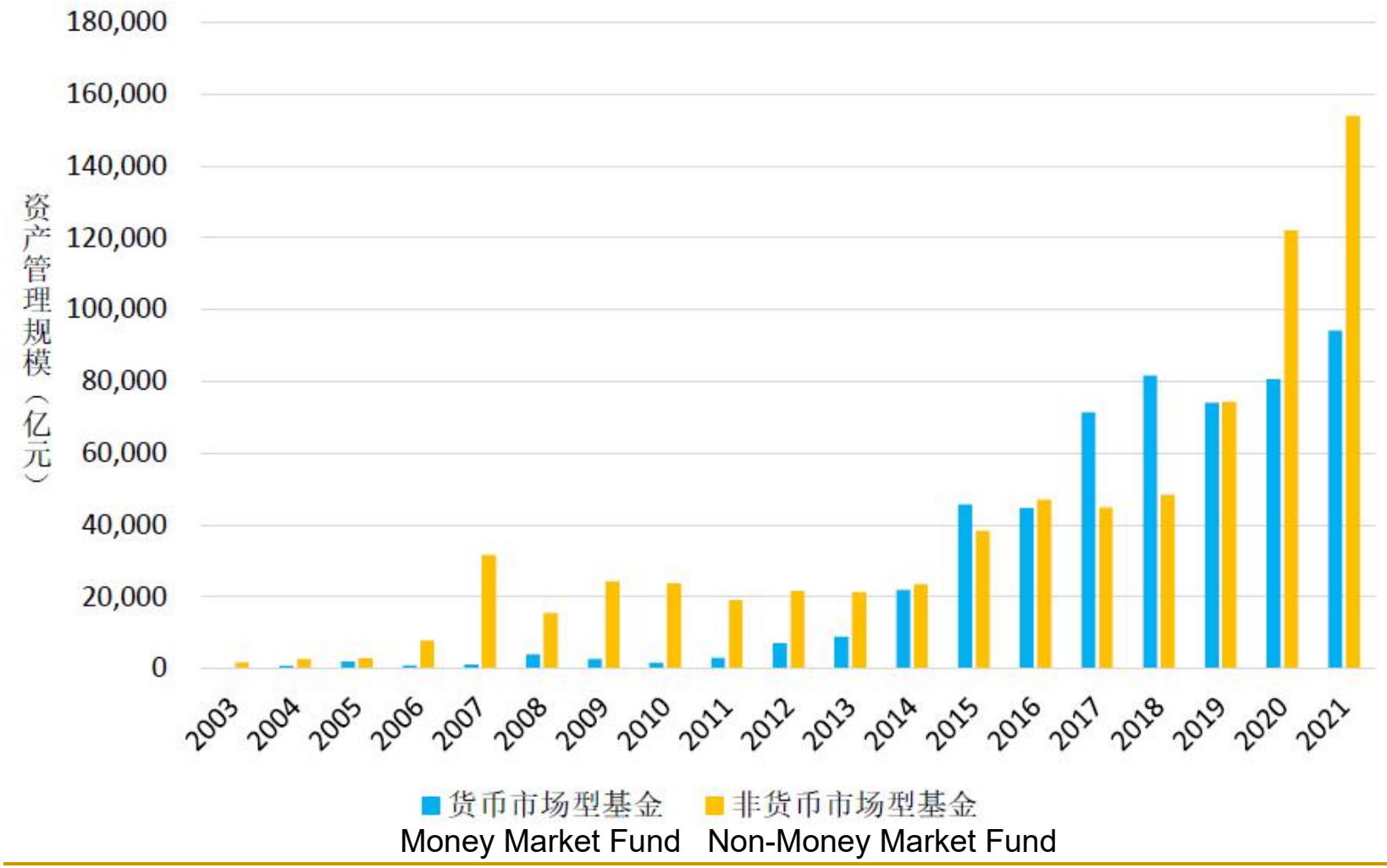


Deposit Bond Fund Stock P&I Others

Can mutual fund manager beat market (China)

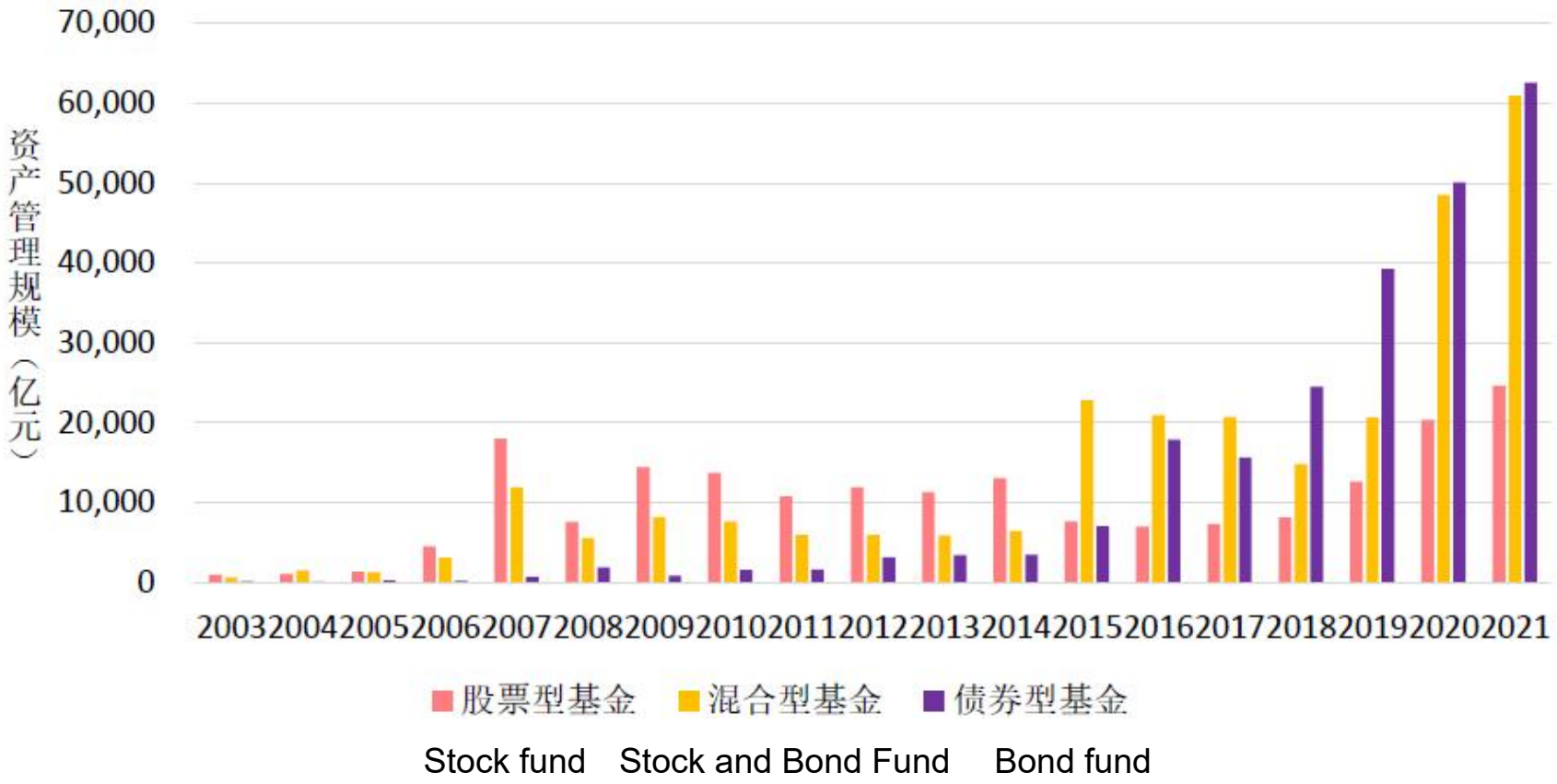
China' GDP in 2021 is 115 trillion Yuan ($17.7 trillion USD)

AUM (100 millions)



Money Market Fund Non-Money Market Fund

Can mutual fund manager beat market (China)



AUM (100 millions)

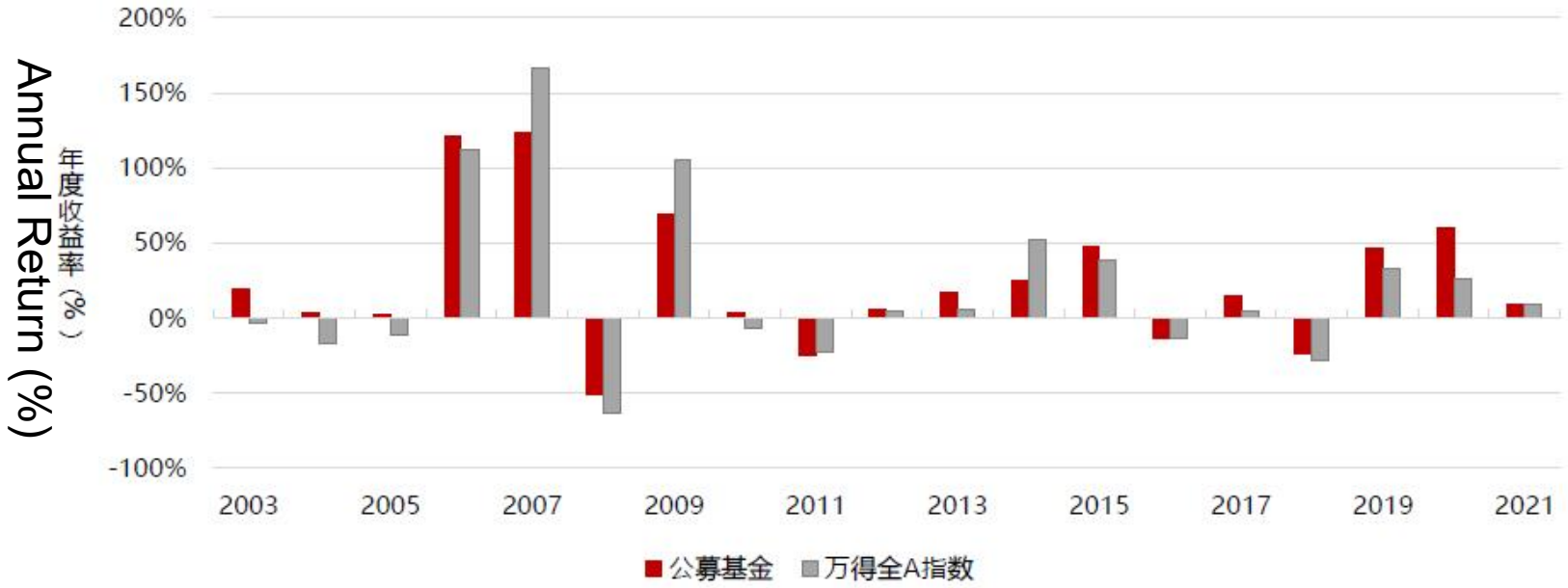
China' GDP in 2021 is 115 trillion Yuan ($17.7 trillion USD)

How to evaluate the performance of China's mutual funds?

- Total return
- Risk adjusted return
  - Sharpe ratio
  - Risk adjusted returns (APT model)
- Performance persistency

Can mutual fund managers beat the market (China)?

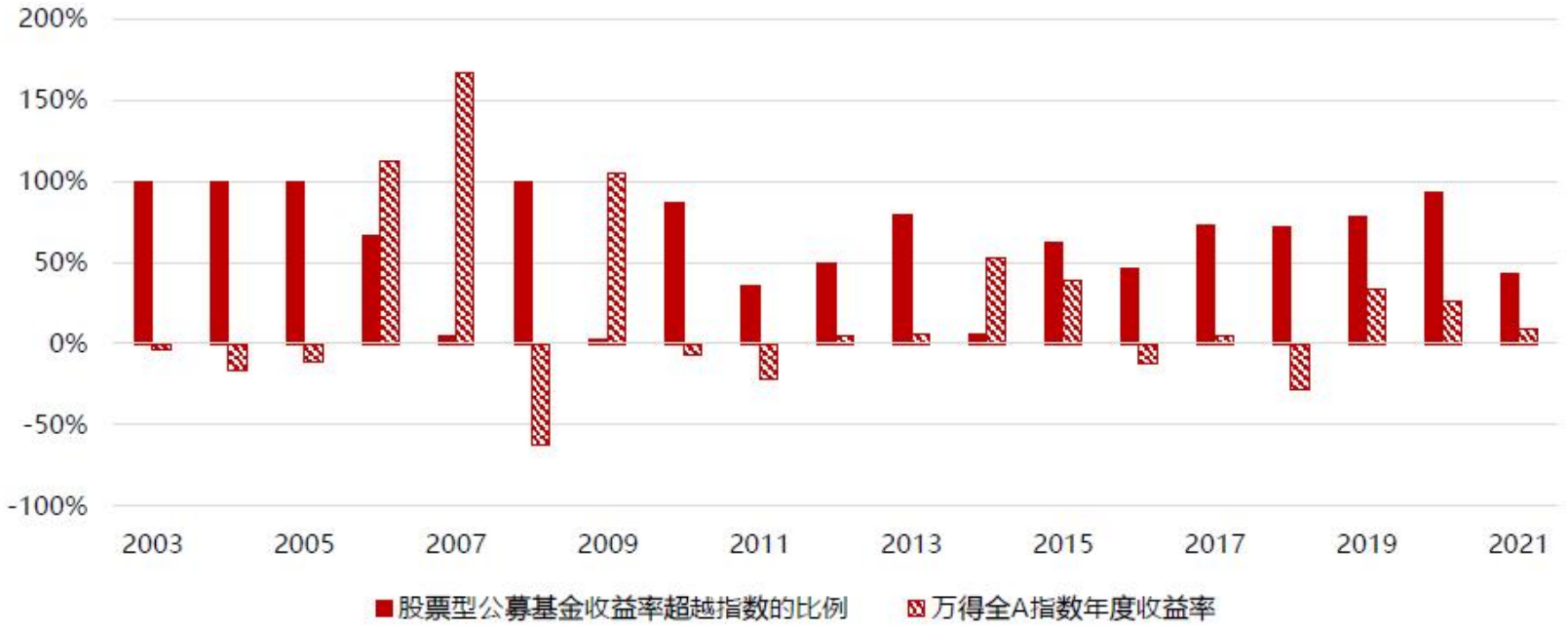
- Mutual fund return is equal weighted



Stock mutual fund Wind China stock market index

Can mutual fund managers beat the market (China)?

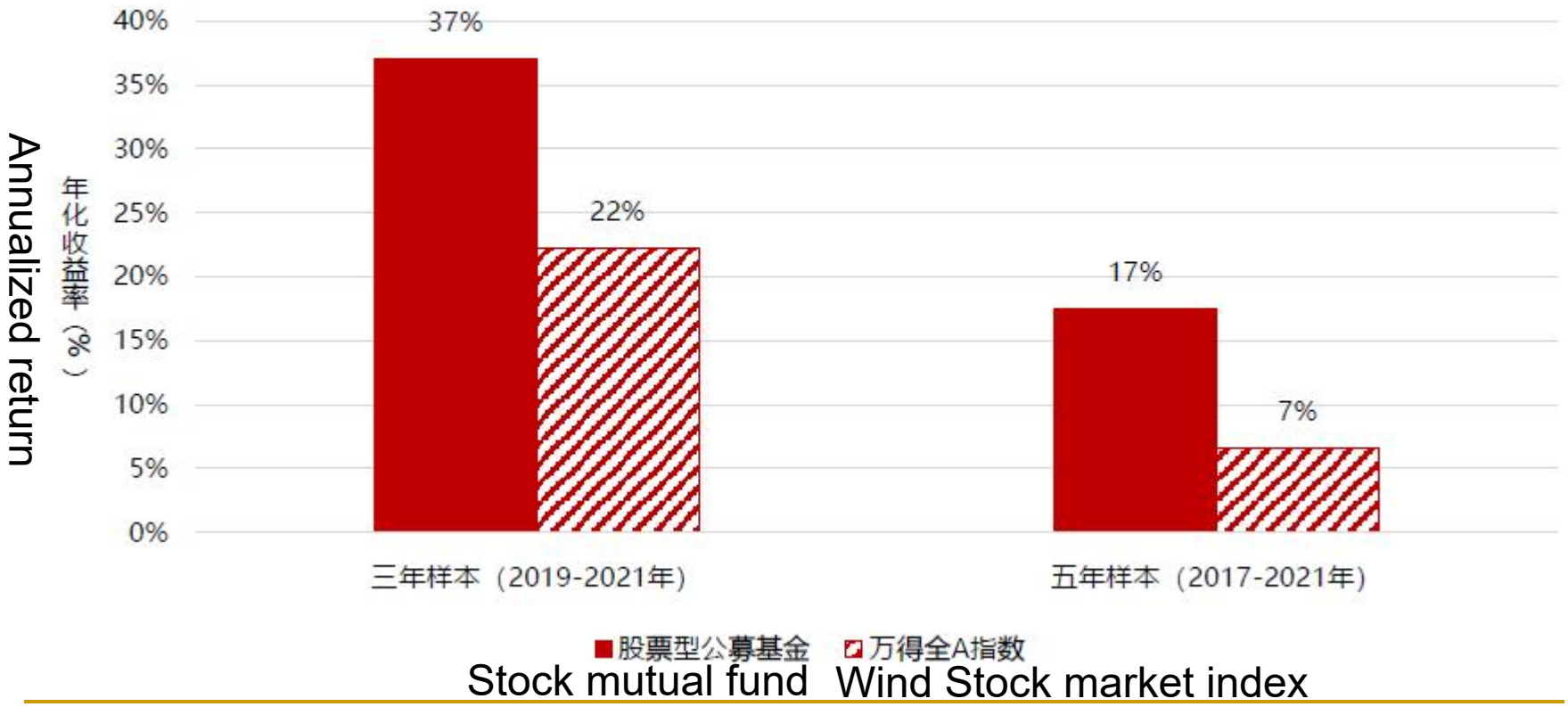
- The percentage of mutual funds to beat the market



The percentage of funds to beat the market Market index return

Can mutual fund managers beat the market (China)?

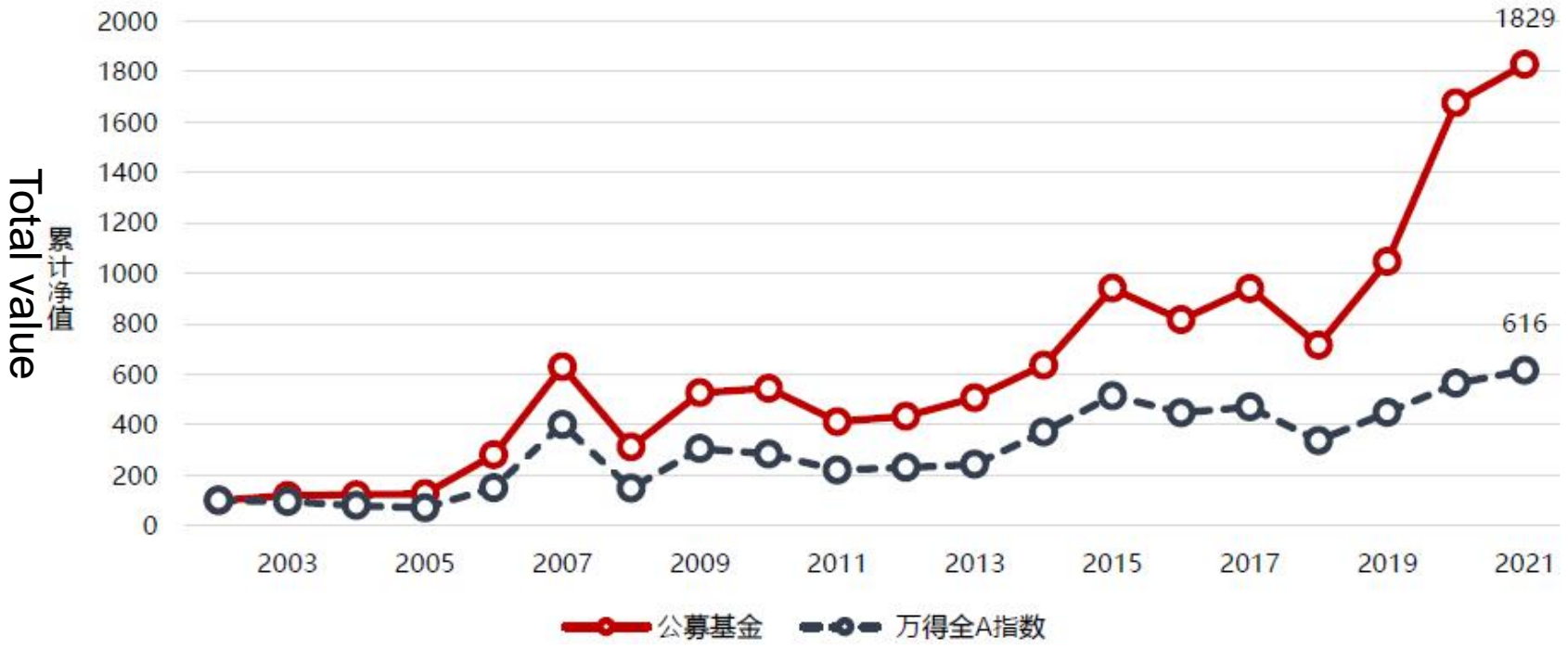
- Cumulative return



Stock mutual fund Wind Stock market index

Can mutual fund manager beat market (China)?

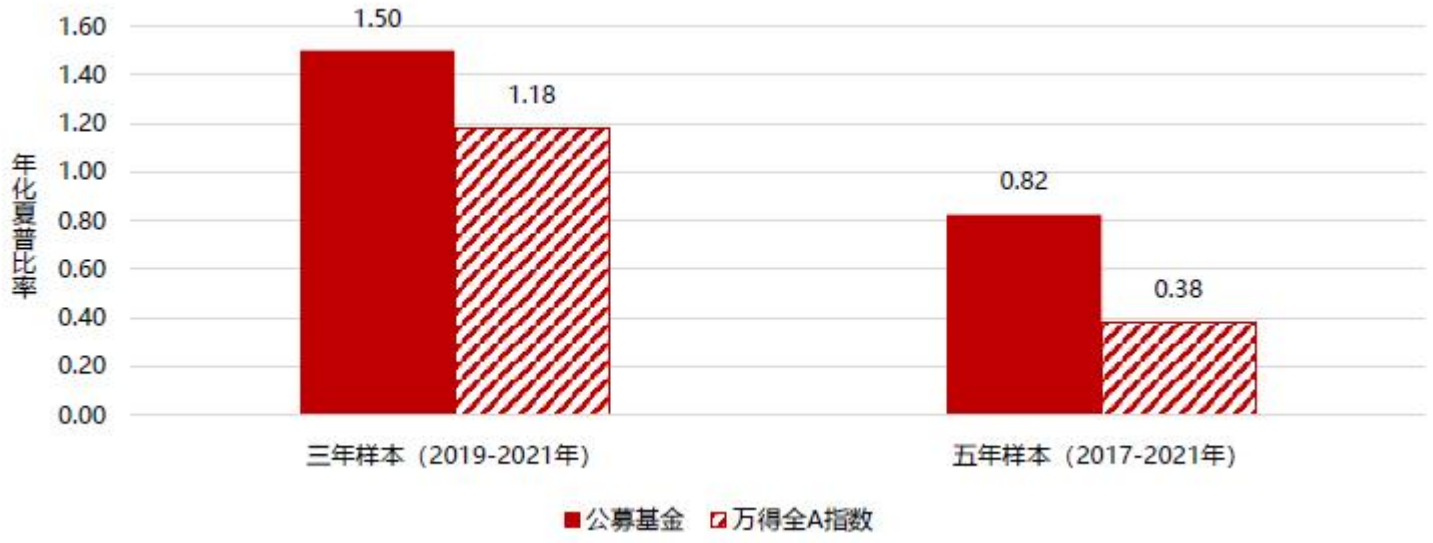
- Cumulative return (Index 10% vs fund 17%)



Stock mutual fund Wind Stock market index

Can mutual fund manager beat market (China)?

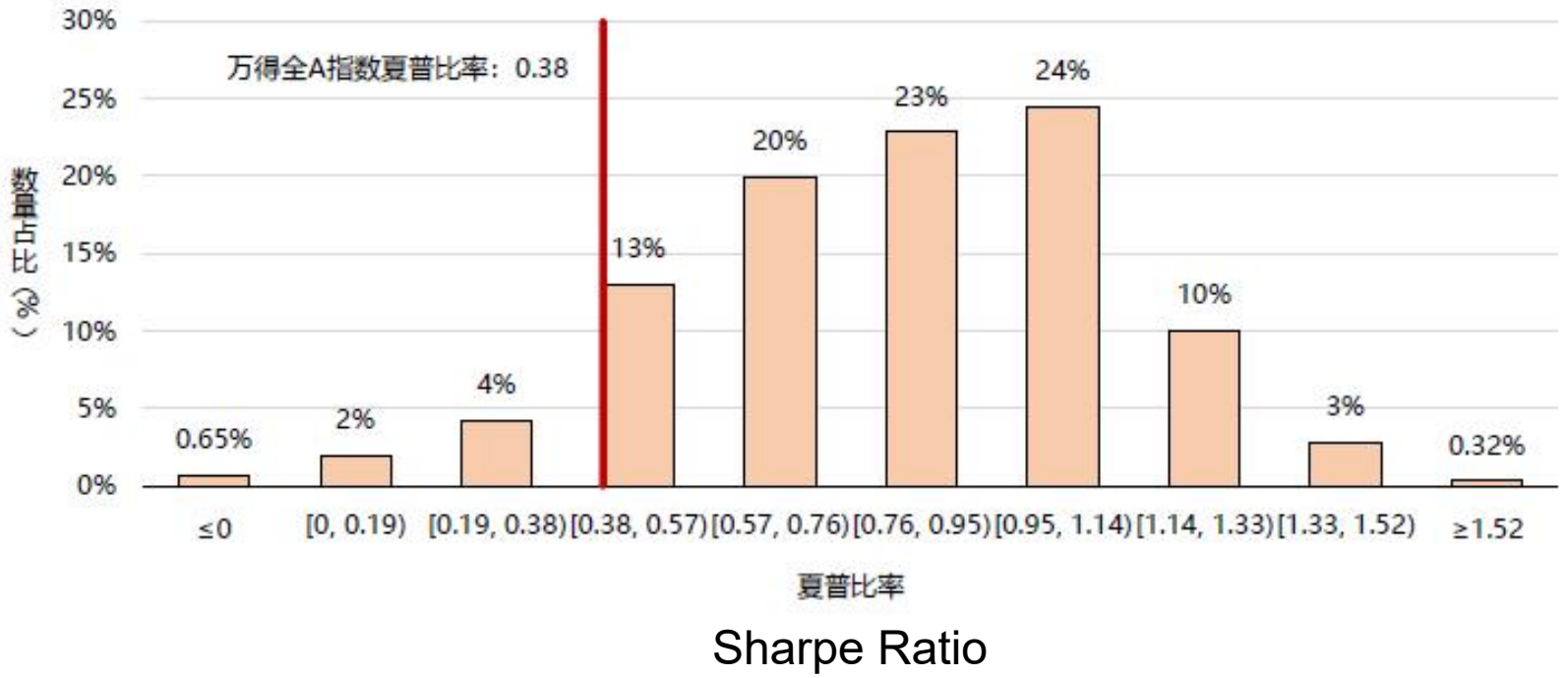
- Sharpe ratio



Stock mutual fund Wind Stock market index

Can mutual fund manager beat market (China)

- The distribution of Sharpe ratio



Sharpe Ratio

Overall

- Mutual fund can beat the market in China
  - Larger average return (on average beat the market by 4%-5% each year)
  - Larger cumulative return
  - In terms of lower Sharpe ratio
  - Then, how?
    - Market timing?
    - Stock picking?

Stock picking capacity

- Whether there is alpha depends on the factor model you use (2006-2020)

表 1　样本基金投资组合回归结果　　单位：%

	等权重组合			资产总值加权组合		
	CAPM	三因子	五因子	CAPM	三因子	五因子
常数项	0.454**	0.452***	0.293*	0.308	0.339**	0.177
	(2.15)	(2.64)	(1.69)	(1.48)	(2.04)	(1.06)
风险溢价因子	79.589***	77.431***	79.973***	80.582***	78.618***	81.244***
	(31.27)	(37.66)	(37.00)	(32.00)	(39.34)	(38.99)
市值因子		2.372	21.632***		-1.886	18.201***
		(0.68)	(3.09)		(-0.56)	(2.70)
账面市值比因子		-47.928***	-35.498***		-52.761***	-39.893***
		(-8.14)	(-5.07)		(-9.22)	(-5.91)
盈利能力因子			21.286*			23.060**
			(1.97)			(2.21)
投资模式因子			-17.409*			-18.139*
			(-1.71)			(-1.84)
样本数量	180	180	180	180	180	180
回归平方和	0.845	0.901	0.907	0.851	0.908	0.915

注：***、**、*分别表示在 1%、5%和 10%的水平下通过显著性检验。括号内为标准差。

Stock picking capacity

- Fama-French four factors model
  - The distribution of alphas

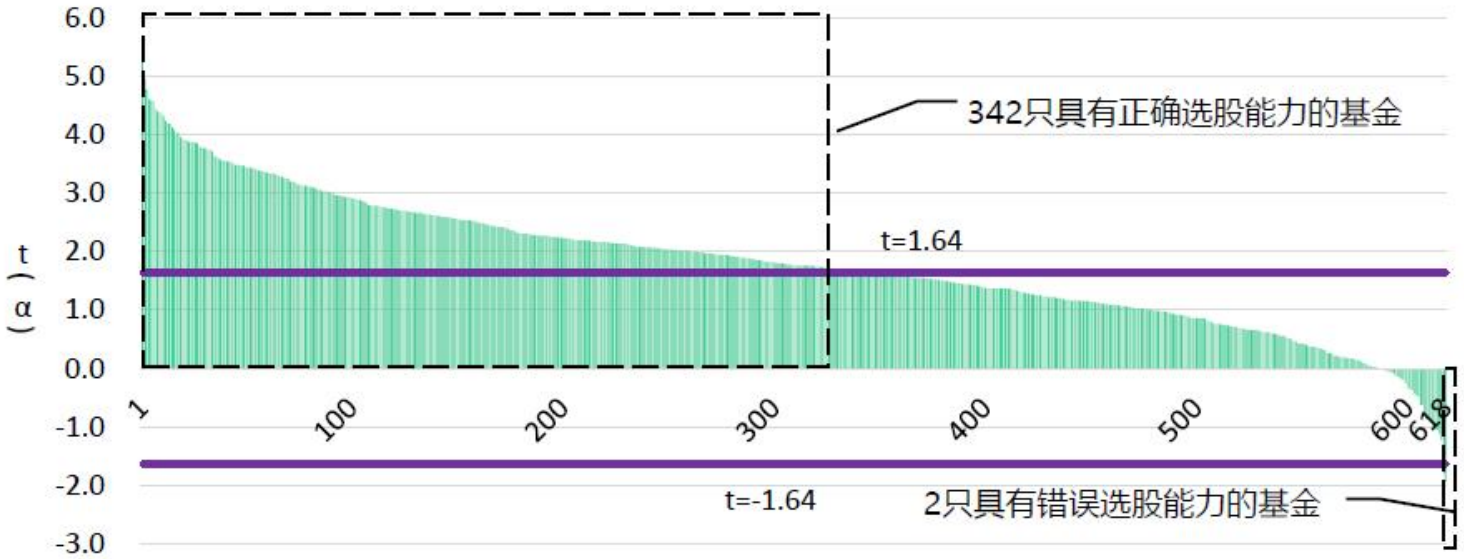


图 3-1　股票型基金的选股能力 α 的 t 值（显著性）排列：2017~2021 年

注：正确选股能力代表 $t(\alpha) > 1.64$；错误选股能力代表 $t(\alpha) < -1.64$；未表现出选股能力代表 $-1.64 \le t(\alpha) \le 1.64$。基金具有选股能力是指基金表现出正确的选股能力，基金不具有选股能力代表基金表现出错误的或未表现出选股能力。

What is market timing?

- Market timing can be profitable theoretically

The Allure of Market Timing: Missing the Worst Days
20 Years (1/1/1995 - 12/31/2014)

$10,000 Invested in the S&P 500 Index	S&P 500 Annualized Return	Value of $10,000 at the end of the period	Gain/ Loss	Impact of Missing Days
All 5,036 trading days	9.85%	$65,475	$55,475	--
Less the 5 days with the biggest losses	12.24%	$100,688	$90,688	63.48%
Less the 10 days with the biggest losses	14.13%	$140,670	$130,670	135.55%
Less the 20 days with the biggest losses	17.19%	$238,681	$228,681	312.22%
Less the 40 days with the biggest losses	22.19%	$550,011	$540,011	873.43%

What is market timing?

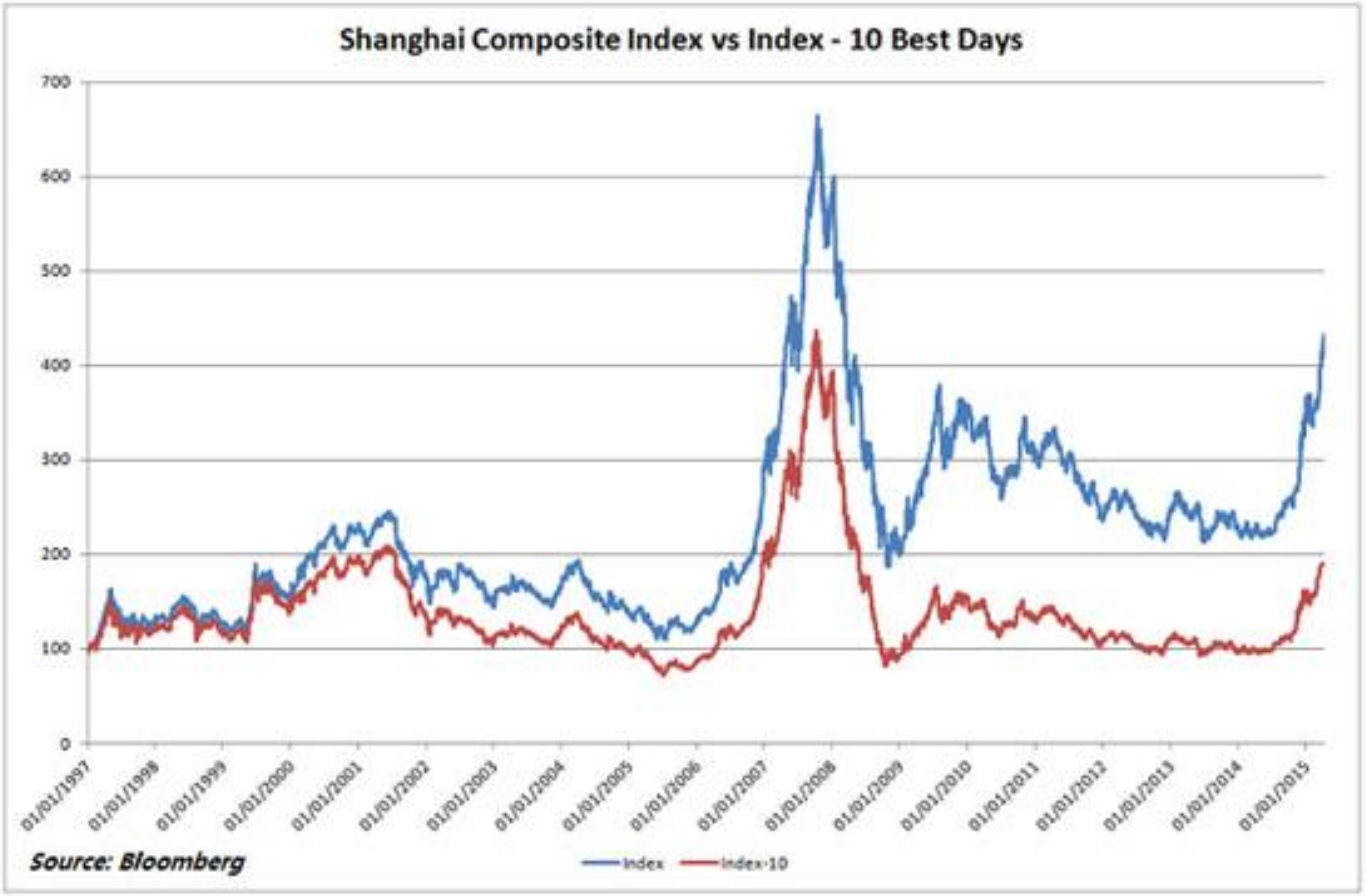
- But, if the market pricing is wrong….

The Problem with Market Timing: Missing the Best Days
20 Years (1/1/1996 - 12/31/2015)

$10,000 Invested in the S&P 500 Index	S&P 500 Annualized Return	Value of $10,000 at the end of the period	Gain/ Loss	Impact of Missing Days
All 5,036 trading days	8.19%	$48,249.94	$38,249.94	--
Less the 5 days with the biggest gains	5.99%	$32,008.68	$22,008.68	-42.46%
Less the 10 days with the biggest gains	4.49%	$24,079.67	$14,079.67	-63.19%
Less the 20 days with the biggest gains	2.05%	$15,004.35	$5,004.35	-86.92%
Less the 40 days with the biggest gains	-1.96%	$6,734.45	-$3,265.55	-108.54%

What is market timing?

- If you miss ten days of the highest return in China?



在这20年间，只要错过这10天，投资（投机）者的回报就只是原来A股指数策略(buy and hold)的回报的一半左右

How to test fund managers' market timing capacity?

- Treynor-Mazuy Model
  - If the coefficient of the square term is positive and significant, it indicates the existence of market timing

$$R_{it} - R_{ft} = \alpha_i + \beta_{im} * (R_{mt} - R_{ft}) + \gamma_i * (R_{mt} - R_{ft})^2 + \beta_{ismb} * SMB_t + \beta_{ihml} * HML_t + \beta_{imom} * MOM_t + \varepsilon_{it}$$

The Market Timing capacity

- The proposition of mutual funds having market timing capacity



Funds with market timing capacity　　Funds without market timing capacity

Can mutual fund manager beat the stock market?

- There is evidence in support of fund managers' stock picking capacity
- There is no evidence in support of fund managers' market timing capacity

How can fund managers beat market in China?

- Individual investors?
  - 从交易量来看，中国的散户的交易量大约占到了市场总交易量的80%左右，而机构和法人的交易量加起来只是占到剩下的20%。所以中国股市的特点就是，占有股份不多的小股东，以非常高的交易频率在股市里买进卖出。
  - 2018年全部美股的换手率为116%，远低于A股的428%。
  - “不少投资者选购股票的认真程度竟不如日常的消费购物，如果把购物时货比三家的认真用于选股，会节省更多的钱.”—彼得林奇
  - 这种特点导致我们的股市可能会有很多噪音（Noise），即脱离股票基本面的无意义波动。比如去年11月，在媒体报道特朗普逐渐取得大选领先时，川大智胜逆市大涨6%。2014年，文章出轨，伊利股份大跌4%。2013年，李天一被抓，ST天一跌停。2012年，奥巴马连任美国总统，澳柯玛涨停。这样的例子还有很多。
- Insider trading?
  - 2017年01月，中国私募基金泽熙投资管理的创始人徐翔被判处有期徒刑五年六个月同时并处罚金。据媒体报道，法院经审理查明，2010年至2015年，被告人徐翔单独或伙同被告人王巍、竺勇，先后与十三家上市公司的董事长或实际控制人，合谋控制上市公司择机发布“高送转”方案、引入热点题材等利好消息；徐翔、王巍基于上述信息优势，使用基金产品及其控制的证券账户，在二级市场进行涉案公司股票的连续买卖，拉抬股价。徐翔组织实施了全部13起证券交易操纵行为，从中非法获得巨额利益。

Can mutual fund managers beat market? (China)

- Is past winner still the winner in the future?
- Weak persistency

%	G1(Lowest)	G2	G3	G4	G5(Highest)	Last year return	Next year return
G1(Lowest)	23.6	18.8	18.3	20.1	19.1	5.01%	24.15%
G2	16.9	22.7	20.4	20.6	19.1	15.6%	25.34%
G3	16.8	17.0	20.8	25.8	19.7	22.25%	26.82%
G4	20.8	22.2	20.9	18.4	17.4	29.26%	24.28%
G5(Highest)	22.9	19.4	18.9	15.1	23.6	42.85%	26.02%

Stock market

- Financial market is like an ocean: mysterious, violent and tempting





Masters' suggestion



If I have noticed anything over these 60 years on Wall Street, it is that people do not succeed in forecasting what's going to happen to the stock market.

- Benjamin Graham

Interview with Hartman Butler, "An hour with Mr. Graham", 1976



There are two kinds of investors, be they large or small: those who don't know where the market is headed, and those who don't know that they don't know. Then again, there is a third type of investor, whose livelihood depends upon appearing to know.

-- William Bernstein

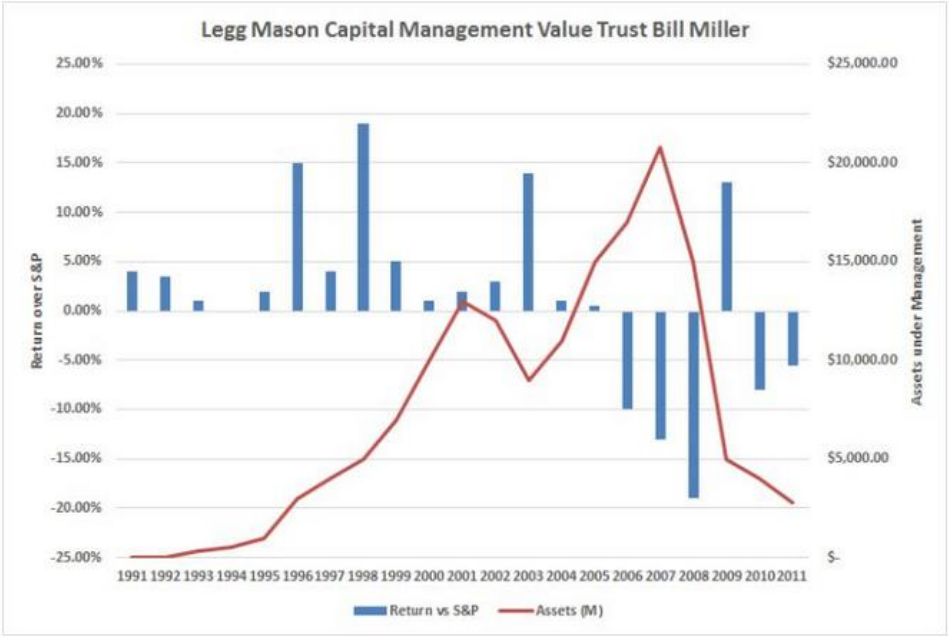
The Intelligent Asset Allocator , 2000

这个世界上有两种投资人：第一种是不知道股市往哪里走的，第二种是不知道他们自己不知道股市的走向的。但是事实上还有第三种人：他们靠假装可以预测股市的走向来骗吃骗喝.

中国的基金经理能否战胜市场？

- Case 1:比尔米勒
  - 在全世界所有的有公开交易记录的基金经理中，比尔米勒的业绩是最好的之一。比尔米勒管理的美盛价值信托基金（Legg Mason Value Trust）曾经在1991年到2005年连续14年战胜标准普尔500指数，创造了一个“投资神话”。



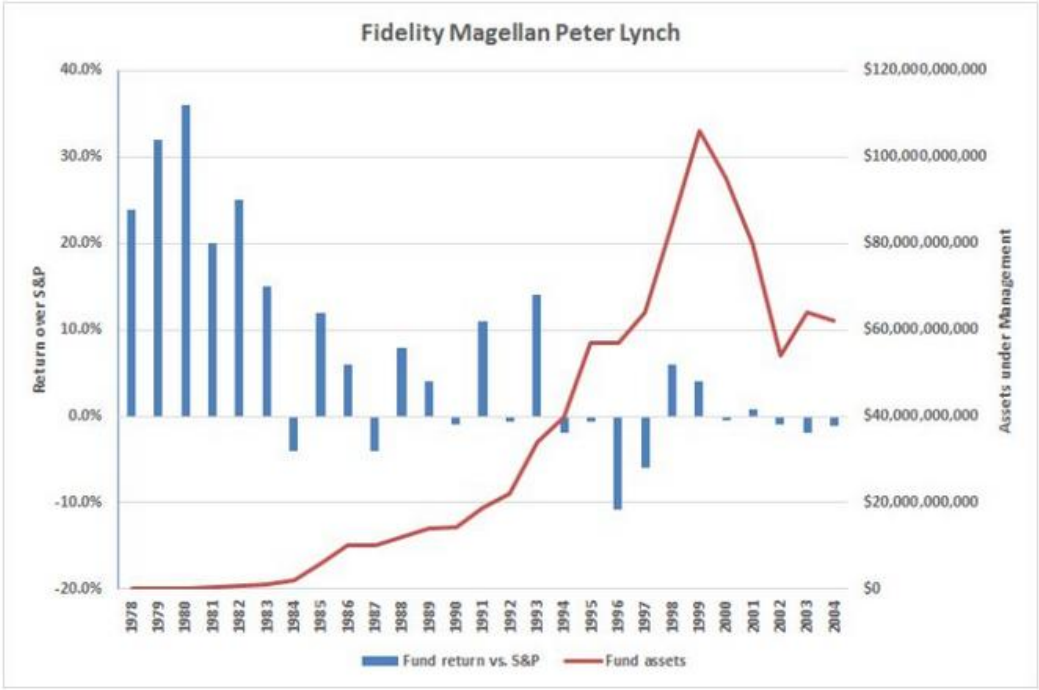


美盛信托基金的资产在2006/07年左右达到巅峰值200亿美元（上图红线），而该基金在2006年以后的业绩则差强人意。比如该基金在2006到2011年的平均超额收益和基准标准普尔500指数相较，大约为**每年负7.1%**. 在该基金业绩最好的时候（1991-1999年），广大投资者并没有得到什么好处。

中国的基金经理能否战胜市场？

- Case 2: 彼得林奇





林奇先生从1978开始管理该基金，直至1990年退休。在这13年中，林奇先生以平均每年13.3%超过基准（标准普尔500指数，S&P500）的收益，年波动率大约13.3%。成就了投资界的另一个神话。在该基金业绩最好的时候（1978-1984年），广大投资者并没有得到什么好处。而该基金绝大部分的投资者（1995年之后），购买到的是一支非常平庸的基金，回报和标准普尔500指数没什么差别。

林奇选股品味：1、傻名字 2、乏味 3、令人厌烦 4、有庇护的独立子公司 5、乏人关注的潜力股 6、充满谣言的公司 7、大家不想关注的行业 8、增长处于零的行业 9、具保护壁垒的企业 10、消耗性大的消费品 11、直接受惠高技术的客户 12、连其雇员也购买的股票 13、会回购自己股份的公司 14.主营业务单一的公司，认为“多样化恶化”。

中国的基金经理能否战胜市场？

- Case 3：王亚伟-华夏大盘精选
  - 王亚伟平和谦逊，文静敦厚。王亚伟说，一个好的基金经理要修炼到“不以物喜，不以己悲”，要让自己的内心能保持清静，只有把“性格”、“品位”和“思想”这三重心门打造到一定境界，才能抵御住投资和人生中无处不在的致命诱惑。
  - ，“作为基金的管理人，投资者把钱交给你就是要你代替他去思考，你要是随波逐流，缺乏独立思考的精神，就是没有做到尽职尽责。”





小股票，重组概念股，光环效应

中国的基金经理能否战胜市场？

Case 3: 王亚伟

- 第一次认识王亚伟还得从1995年华夏证券东四营业部说起。1995年的东四营业部是全国最大的证券营业部之一，除了传统的股票经纪业务外，还从事投行、自营、研究、期货等业务，相当于一个中等证券公司的规模，当时我任总经理。90年代中期的证券业还是一个新兴的行业，证券从业人员数量少且大部分是从银行转行而来，专业水平与今日不可同日而语。
- 为提高营业部水平需要招收一批优秀人才，但营业部刊发广告很难招聘到优秀人才。听说华夏证券总公司正在招聘人才，我就采用“借鸡下蛋”的办法，对总部人力资源部讲，把不符合总部标准的简历转给我，里面或许有适合营业部工作的人，营业部还可节省一笔广告费。
- 我从一大堆简历中选了5个人。经过电话联系面试，其中4人听说是营业部后都拒绝了。联系王亚伟时，他同意来营业部谈谈。他教育背景不错，在清华大学取得了双学位。
- 面试时，他给我最大的印象是平和、淡定。吸取面试前面4个人的教训，我向他描绘了营业部的宏伟发展蓝图，希望他能来营业部工作。他当时就答应了，也许是缘分吧。后来才得知，王亚伟是1989年安徽省理科高考状元，大学期间就对证券投资产生了兴趣。
- 1995年10月，王亚伟加盟华夏证券东四营业部的研究部从事研究咨询工作。东四营业部的研究部主要负责为客户提供一些日常的投资咨询服务，也就是我们常说的股评，谈不上深入的基本面研究，与现在的证券研究所相距甚远。
- 我发现王亚伟对研究很感兴趣。他没有把工作停留在简单的股评上，而是从企业基本面入手发掘投资机会，为客户推荐了一些好股票。一年后任研究部经理。他为人低调，生活简朴，一家人租住在东四十条半地下的房子里。
- 一天，我对王亚伟说：“我在营业部做股票自营赚了2000多万元还没有进大账，你就接着负责这2000万元的自营账户吧。反正是赚来的钱，尽量不亏损就行了。”或许这就是他的第一个“基金”吧。那几年行情不好，他居然做到了不亏钱还略有小赚。
- 1998年2月，我正在负责组建华夏基金管理公司。我找到王亚伟，问他愿意不愿意到基金公司工作？他憨笑着说：“您觉得行，我就跟您去。”说实在的，基金刚刚试点，将来如何还很难说；况且东四营业部作为国内最大的营业部，年利润近2亿元，被业内称为“华夏第一营业部”，工作环境及收入都很好。王亚伟放弃东四营业部稳定的工作，选择到准备试点的基金公司工作也是需要一点勇气的。

Summary

- Various institutional investors
- No superior performance, especially after fees
- Very little persistence in performance
- Inconclusive implications for market efficiency